Worksheet 5.3 — Derivatives of Trig, Exponential and Log Functions

Mathematics: Analysis & Approaches SL (IB) · Unit 5: Differential Calculus

The Big Ideas

Base derivatives for sin, cos, exp, and ln extend with the chain, product, and quotient rules.

- $\frac{d}{dx}[\sin x] = \cos x$, $\frac{d}{dx}[\cos x] = -\sin x$, $\frac{d}{dx}[e^x] = e^x$, $\frac{d}{dx}[\ln x] = \frac{1}{x}$.

Worked Examples

Example 1 — Chain rule with exponential

Differentiate $f(x) = e^{4x^2-x}$.

Solution. Inner derivative $8x - 1$.

Conclusion: $f'(x) = (8x-1)e^{4x^2-x}$.

Example 2 — Chain rule with logarithm

Differentiate $f(x) = \ln(7x^2+3)$.

Solution. Inner derivative $14x$.

Conclusion: $f'(x) = \frac{14x}{7x^2+3}$.

Example 3 — Product rule with exponential

Differentiate $f(x) = x^2e^{3x}$.

Solution. $2xe^{3x} + x^2(3e^{3x}) = xe^{3x}(2+3x)$.

Conclusion: $f'(x) = xe^{3x}(3x+2)$.

Example 4 — Deriving sec x

Using the quotient rule and $\sec x = \frac{1}{\cos x}$, prove $\frac{d}{dx}[\sec x] = \sec x \tan x$.

Solution. $\frac{0 \cdot \cos x - 1 \cdot (-\sin x)}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}$.

Conclusion: $= \sec x \tan x$.

Example 5 — Chain rule with sine of a linear function

Differentiate $f(x) = \sin(5x-2)$.

Solution. Inner derivative is 5.

Conclusion: $f'(x) = 5\cos(5x-2)$.

Example 6 — Chain rule nested inside a trig function

Differentiate $f(x) = \cos(x^2 + 3x)$.

Solution. Inner derivative is $2x + 3$.

Conclusion: $f'(x) = -(2x + 3)\sin(x^2 + 3x)$.

Example 7 — Product rule combining trig and log

Differentiate $f(x) = \sin x \ln x$.

Solution. $u = \sin x, v = \ln x$: $f'(x) = \cos x \ln x + \sin x \cdot \dfrac{1}{x}$.

Conclusion: $f'(x) = \cos x \ln x + \dfrac{\sin x}{x}$.

Example 8 — Quotient rule with exponential and trig

Differentiate $f(x) = \dfrac{e^x}{\cos x}$.

Solution. $\dfrac{e^x \cos x - e^x(-\sin x)}{\cos^2 x} = \dfrac{e^x(\cos x + \sin x)}{\cos^2 x}$.

Conclusion: $f'(x) = \dfrac{e^x(\cos x + \sin x)}{\cos^2 x}$.

Example 9 — Application — rate of a decaying quantity

A radioactive sample's mass is $M(t) = 200e^{-0.05t}$ grams. Find the rate of change of mass at $t = 10$.

Solution. $M'(t) = 200(-0.05)e^{-0.05t} = -10e^{-0.05t}$; $M'(10) = -10e^{-0.5}$.

Conclusion: $M'(10) \approx -6.07$ g/unit time — the mass is decreasing at about 6.07 g per unit time.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Differentiate $f(x) = 7\cos x - e^x + 3\ln x$.

Question 2

Differentiate $f(x) = e^{6x-1}$.

Question 3

Differentiate $f(x) = \ln(4x^3 - x)$.

Question 4

Differentiate $f(x) = x^3 \sin x$ using the product rule.

Question 5 — Challenge

Differentiate $f(x) = \dfrac{e^x}{x^2}$ using the quotient rule.

Question 6 — Challenge

Using the quotient rule and $\csc x = \dfrac{1}{\sin x}$, derive $\dfrac{d}{dx}[\csc x] = -\csc x \cot x$.

Question 7

Differentiate $f(x) = \sin(3x + 1)$.

Question 8

Differentiate $f(x) = \cos(x^2 - 4x)$.

Question 9

Differentiate $f(x) = \cos x \ln x$ using the product rule.

Question 10 — Challenge

Differentiate $f(x) = \dfrac{e^x}{\sin x}$ using the quotient rule.

Question 11

A sample's mass is $M(t) = 150e^{-0.03t}$ grams; find the rate of change at $t = 20$.

Question 12

Differentiate $f(x) = e^{\sin x}$.

Question 13 — Challenge

Differentiate $f(x) = \ln(\cos x)$, stating the domain restriction this creates.

Question Recap & Answer Key

Question Recap

1. Differentiate $f(x) = 7\cos x - e^x + 3\ln x$.
2. Differentiate $f(x) = e^{6x-1}$.
3. Differentiate $f(x) = \ln(4x^3 - x)$.
4. Differentiate $f(x) = x^3 \sin x$ using the product rule.
5. Differentiate $f(x) = \dfrac{e^x}{x^2}$ using the quotient rule.
6. Using the quotient rule and $\csc x = \dfrac{1}{\sin x}$, derive $\dfrac{d}{dx}[\csc x] = -\csc x \cot x$.
7. Differentiate $f(x) = \sin(3x + 1)$.
8. Differentiate $f(x) = \cos(x^2 - 4x)$.
9. Differentiate $f(x) = \cos x \ln x$ using the product rule.
10. Differentiate $f(x) = \dfrac{e^x}{\sin x}$ using the quotient rule.
11. A sample's mass is $M(t) = 150e^{-0.03t}$ grams; find the rate of change at $t = 20$.
12. Differentiate $f(x) = e^{\sin x}$.
13. Differentiate $f(x) = \ln(\cos x)$, stating the domain restriction this creates.

✓ Answer Key

1. $-7\sin x - e^x + \dfrac{3}{x}$
2. $6e^{6x-1}$
3. $\dfrac{12x^2 - 1}{4x^3 - x}$
4. $3x^2 \sin x + x^3 \cos x$
5. $\dfrac{e^x(x^2) - e^x(2x)}{x^4} = \dfrac{e^x(x-2)}{x^3}$
6. $\dfrac{0 \cdot \sin x - 1 \cdot \cos x}{\sin^2 x} = -\dfrac{\cos x}{\sin^2 x} = -\dfrac{1}{\sin x} \cdot \dfrac{\cos x}{\sin x} = -\csc x \cot x$
7. $3\cos(3x + 1)$
8. $-(2x - 4)\sin(x^2 - 4x)$
9. $-\sin x \ln x + \dfrac{\cos x}{x}$
10. $\dfrac{e^x \sin x - e^x \cos x}{\sin^2 x} = \dfrac{e^x(\sin x - \cos x)}{\sin^2 x}$
11. $M'(t) = -4.5e^{-0.03t}$; $M'(20) = -4.5e^{-0.6} \approx -2.47$ g/unit time
12. $\cos x \cdot e^{\sin x}$
13. $\dfrac{-\sin x}{\cos x} = -\tan x$; domain requires $\cos x > 0$, so excludes wherever $\cos x \leq 0$